

A Novel Compliant Nanopositioning Stage Driven by a Normal-Stressed Electromagnetic Actuator

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Abstract—This article reports on the design, modeling, control, and testing of a novel compliant nanopositioning stage driven by a self-developed normal-stressed electromagnetic actuator. To facilitate the parameter selection for the stage to achieve the desired stroke and natural frequency, an analytical model of both the electromagnetic circuit and flexure mechanism is established, which is then systematically verified through finite element analysis. By combining a proportional—integral—differential (PID)-based main controller with a system dynamics inversion-based feedforward compensator, a closed-loop control system for the stage is constructed with the main controller being tuned through Bode's ideal transfer function-based loop-shaping method. The experimental result shows that a stroke of $\pm 95 \mu\text{m}$ and a first natural frequency of 743 Hz are achieved for the designed stage. Finally, taking advantage of the constructed control system, the nanopositioning capability is demonstrated by finely tracking the harmonic and nanostair commands.

Note to Practitioners—Piezoelectric actuators (PEAs) and voice coil motors (VCMs) are commonly adopted for the actuation of nanopositioning stages. In general, constrained by the inherent small strain, PEA is more suitable for the generation of dynamic motions in dozens of micrometers. Meanwhile, VCM has a theoretically infinite motion range but a slow response due to its low force density nature. Taking advantage of a newly developed normal-stressed electromagnetic actuator (NSEA), we demonstrate the design, modeling, control, and testing of an NSEA-based nanopositioning stage with a relatively long motion range, high bandwidth, and compact size. Considering the high force density and relatively large motion range only constrained by the effective air gap, the NSEA-based stage is demonstrated herein to be very promising to achieve highly dynamic motions within hundreds of micrometers.

Index Terms—Compliance modeling, flexure mechanism, fractional order control, nanopositioning stage, normal-stressed electromagnetic actuator (NSEA).

I. INTRODUCTION

NANOPOSITIONING stages have been applied in a wide spectrum of fields, including the micro/nanomanufacturing [1], [2], micro/nanomanipulation [3], and scanning probe based nanoimaging [4], to mention a few. Currently, the actuators can be mainly classified according to the principle for force generation into the smart material-based [2], [3] and the electromagnetic actuators [5]–[7].

With the smart material-based actuators, the piezoelectric actuator (PEA) might be the most popular one, which has been widely applied in both academia and industry due to its outperforming properties, including the compact size, high stiffness, high force density, and high motion resolution, to mention a few [8]–[10]. However, the inherently small strain rate may lead to limited motion strokes, mainly in dozens of micrometers [6]. To overcome this limitation, a variety of compliant amplifiers were designed to amplify motions for the PEA [11]–[13]. Nevertheless, complicated and multistage amplifier mechanisms may significantly increase the structure size and deteriorate the system's dynamic response. Therefore, the PEA is commonly regarded as more suitable for the applications requiring high dynamic responses but short strokes, for example, the high-speed scanning for the atomic force microscopy [4], [14] and the fast tool oscillation for the micro/nanocutting [15], [16]. With the electromagnetically actuated nanopositioning stages, the most popular driving source might be the shear-stressed Lorentz force [6], [7]. Although the Lorentz force-based actuation features high linearity and unlimited motion stroke in theory, its inherently low force density makes it unsuitable for high bandwidth applications, and the over-large size is usually required to gain desired actuation behavior. On basis of these features, the Lorentz force-based actuator is more commonly for long travel actuation, ranging from hundreds of micrometers to dozens of millimeters with low working bandwidth [6], [17].

In micromanufacturing, an alternative electromagnetic actuation was developed on basis of the normal-stressed electromagnetic actuator (NSEA), which had a much higher force density than the Lorentz force actuator, and full utilization of the armature surface area was achieved by adopting a *E*-shaped magnetic circuit [18]. Taking advantage of a bias magnetic flux in NSEA, a linear relationship between the actuation force and

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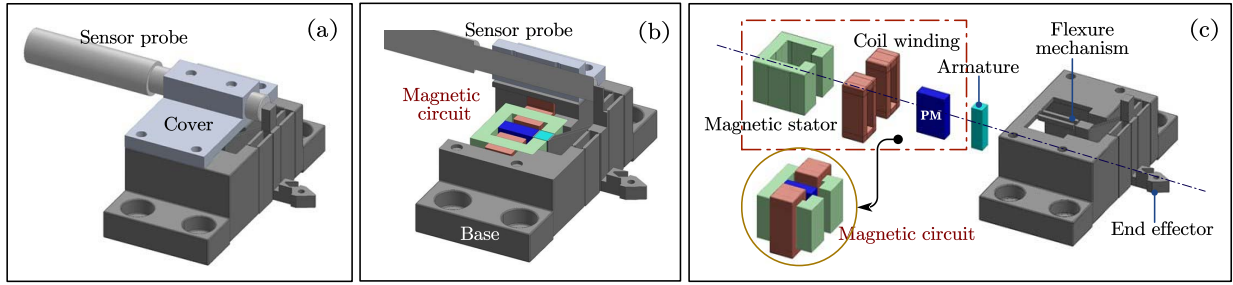


Fig. 1. Structure of the stage, (a) 3-D assembled model, (b) partial cross section view, and (c) explosive view without the sensor and cover.

the driving current can be achieved, which is very promising for the high-precision motion control [18], [19]. Recently, the NSEA is mainly developed to apply to actuate the cutters in micromanufacturing, for example, to construct fast tool servos for the ultraprecision turning of complex-shaped surfaces [18], [20] and provide external forces to stabilize the chatter in cutting [21]. Moreover, it was also combined with an air-bearing spindle to serve as an axial bearing/actuator for the construction of a rotary-axial spindle for the machining [5]. Essentially, the travel range of NSEA is limited by the effective working air gap between the armature and stator, which was demonstrated to be capable of generating motions ranging from several micrometers [18], [22] to millimeters [19].

To deliver the linear motions for the NSEA, the stator and armature are usually manufactured separately and assembled together with tight tolerances, requiring very high manufacturing accuracy. Similar to other nanopositioning stages, the flexure mechanisms are normally used to support the end-effector. Nevertheless, the flexure mechanisms in the prior designs for the NSEA were simple without the careful consideration of the parasitic motion errors. For example, for fast tool servos developed in [18] and [22], a rubber pad was glued between the stator and armature as a special bearing. Since the stator was placed in front of the armature to construct the working air gap, a long shaft had to be utilized to transit the motions to the cutter outside of the actuator, and the shaft was then supported by a monolayer flexure hinge. With this design, the unconstrained tilting degree-of-freedom may lead to large motion errors when subjecting to external loads, especially for the over-hanged shaft [18]. Moreover, with the large-stroke NSEA-based stage traveling in the millimeter range [19], there was no consideration for the suppression of the parasitic motions, which may lead to relatively large error motions along the direction vertical to the actuation axis. In short, although the NSEA is demonstrated to be very promising to achieve a relatively large stroke with a high force density, very limited attention was paid to the design of the NSEA-based stages for the high-performance nanopositioning. Moreover, the powerful flexure mechanisms remained unexplored to extend the positioning capability of the NSEA-based stages for the current designs.

In this article, a novel NSEA-based nanopositioning stage is developed and demonstrated to be outperforming, for which a new flexure mechanism is designed to work with a self-developed NSEA to simultaneously adjust the motion direction and amplify the motion stroke. Facilitated by the established analytical model, the procedure for deter-

mining both the electromagnetic and mechanical parts is detailed, which may provide a solid basis for the design of NSEA-based nanopositioning stages. Finally, Bode's ideal transfer function-based loop-shaping method is developed to design the stage control system to achieve the nanoaccuracy positioning.

II. NSEA-BASED NANOPositioning STAGE

To overcome the limited strain of the PEAs and the limited force density of the Lorentz-force-based voice coil motors, an NSEA is developed to achieve a medium-range motion (hundreds of micrometers) with relatively high bandwidth. Since the actuation force for the NSEA originates from the attraction between the armature and the stator, a special flexure mechanism is required to transit the attraction force to the end-effector, which moves perpendicularly to the attraction direction. To fulfill this requirement, an NSEA-based stage with a deliberately designed flexure mechanism is developed with the assembled structure illustrated in Fig. 1(a). In addition, to better show the inner structure of the stage, a partial cross section view of the assembled structure is presented in Fig. 1(b), and an explosive view without the sensor probe and the cover is further shown in Fig. 1(c).

As shown in Fig. 1, a permanent magnet (PM) and a stator armed by the serially connected coil windings construct the magnetic circuit to drive an armature, forming the basic NSEA unit. The flexure mechanism is designed to support the NSEA and deliver the attractive force on the armature to the end-effector of the stage.

A. Normal-Stressed Electromagnetic Actuator

The equivalent magnetic circuit for the NSEA is schematically illustrated in Fig. 2. A C-shaped stator using soft magnetic material is employed to construct the magnetic circuit, and a PM is employed at the middle to provide a dc biasing flux B_d . On two arms of the stator, the serially connected coil winding will carry an excitation current to generate an ac flux B_a for armature actuation.

Since effective areas of the two armature polar surfaces are the same, the generated ac flux passing through each surface might be identical based on the Gaussian law, irrespective to the armature motion x . Therefore, with an initial air gap of d_0 at each side, the total ac flux can be expressed as

$$B_a = B_{a1} = B_{a2} = \frac{NI\mu_0}{2d_0} \quad (1)$$

where μ_0 is the permeability of vacuum, and B_{a1} and B_{a2} stand for the ac flux densities inside the working air gaps.

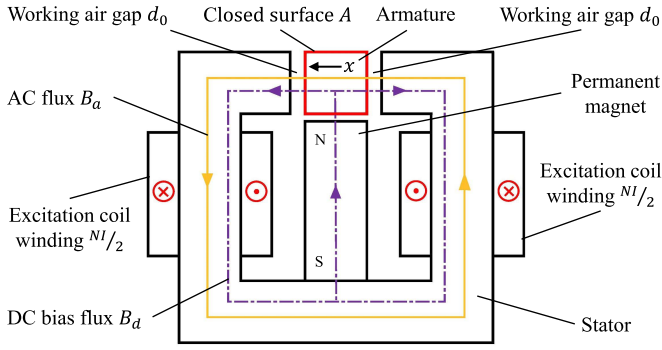
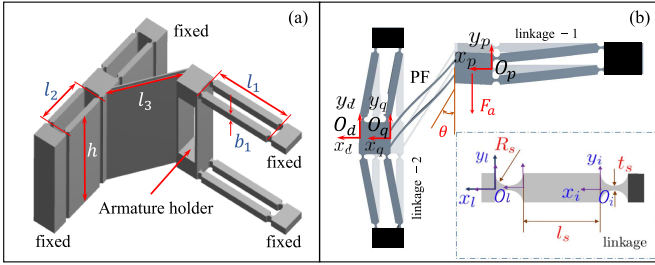


Fig. 2. Principle of the NSEA.

Fig. 3. Structure of flexure mechanism, (a) 3-D model and (b) projected 2-D structure, where l_s is the length of the mid-beam for the linkage.

When the armature has a working distance of x , the dc flux induced by the PM can be obtained inside the two working air gaps as

$$B_{d1} = \frac{d_0 + x}{d_0} B_d, \quad B_{d2} = \frac{d_0 - x}{d_0} B_d \quad (2)$$

where $B_d = \alpha B_r$ with B_r denoting the PM remanence, and α is the magnetic leakage factor.

Therefore, through Maxwell's stress tensor theory, the induced actuation force on the armature can be obtained as [23]

$$\begin{aligned} F_a &= F_1 - F_2 = \frac{(B_{a1} + B_{d1})^2}{2\mu_0} S_a - \frac{(B_{a2} - B_{d2})^2}{2\mu_0} S_a \\ &= \frac{B_d S_a}{d_0} NI + \frac{2B_d^2 S_a}{\mu_0 d_0} x. \end{aligned} \quad (3)$$

As shown in (3), the actuation force is linearly related to the excitation current I and the armature motion x .

B. Mechanical Structure of the Flexure Mechanism

The 3-D structure of the designed flexure mechanism is extracted, as presented in Fig. 3(a), and the 2-D projected view of the mechanism is further shown in Fig. 3(b). As an enabled component for the stage, a simple P: prismatic, and R: revolute (PRRP) mechanism [24], [25] [denoted by parallel leaf-type flexure hinge (PF)] having a pair of parallel leaf-spring flexure hinges (LSFH) is employed to connect the input and output end, which may simultaneously contribute to moving direction modification and actuation travel amplification.

As illustrated in Fig. 3, to improve the input stiffness and generate the linear motion at the input end, a single compliant parallelogram mechanism using four parallel linkages (linkage-1) is specially designed to be monolithically manufactured with the frame. Furthermore, a double-parallelogram mechanism using four linkages (linkage-2) is designed to support the output end to suppress undesired parasitic motions. With respect to all the linkages, each one is serially connected by two right circular flexure hinges (RCFHs) and one flexure beam, as shown in Fig. 3(b).

Note: The flexure mechanism and the mechanical base of the stage are specially designed to be fabricated in a monolithic way, which may avoid the complex assembly as for the reported in [18], [21], and [22], so as to release the manufacturing tolerance required for the assembly.

III. MODELING OF THE FLEXURE MECHANISM

Detailed modeling of both the static and dynamic performance related to the structural parameters is conducted for the flexure mechanism. With respect to the modeling, the flexure hinges undergo small elastic deformation, and the geometry and material nonlinearity are assumed to be ignored to simplify the modeling process.

A. Static Modeling of the Mechanism

The matrix based compliance modeling method is adopted to relate mechanism compliance to its structural parameters. With respect to the basic linkage in Fig. 3(b), its compliance C_l can be expressed in its local coordinate system $O_l - x_l y_l z_l$ as

$$C_l = \sum_{i=1,3} \mathbf{T}_{i,l} \mathbf{C}_r (\mathbf{T}_{i,l})^T + \mathbf{T}_{2,l} \mathbf{C}_s (\mathbf{T}_{2,l})^T \quad (4)$$

where $\mathbf{C}_r$ and $\mathbf{C}_s$ denote the complete compliance matrix of the RCFH and LSFH, respectively.

With the axis-symmetric double-parallelogram mechanism supported by four linkages in parallel, its compliance can be expressed in its local coordinate system $O_d - x_d y_d z_d$ as

$$C_d = \left\{ (\mathbf{C}_L)^{-1} + [\mathbf{T}_{ry}(\pi) \mathbf{C}_L \mathbf{T}_{ry}(\pi)^T]^{-1} \right\}^{-1} \quad (5)$$

where $\mathbf{C}_L = \{\sum_{i=1}^2 [\mathbf{T}_{i,d} \mathbf{C}_l (\mathbf{T}_{i,d})^T]^{-1}\}^{-1}$ is the compliance of the left parallelogram mechanism having two linkages, and $\mathbf{T}_{ry}(\pi) = \text{diag}[\mathbf{R}_y, \mathbf{R}_y]$ with $\mathbf{R}_y$ denoting the standard rotational matrix around the y -axis with an angle of π .

With PF mechanism having two parallel LSFHs, its compliance in the local coordinate system $O_q - x_q y_q z_q$ can be expressed by

$$\mathbf{C}_q = \left[\sum_{i=1}^2 (\mathbf{T}_{i,q} \mathbf{C}_s \mathbf{T}_{i,q}^T)^{-1} \right]^{-1} \quad (6)$$

Similarly, the compliance C_p at point o_p contributed by the four linkage-1 linkages supporting the armature can be determined by

$$\mathbf{C}_p = \left[\sum_{i=1}^2 (\mathbf{T}_{i,p} \mathbf{C}_l \mathbf{T}_{i,p}^T)^{-1} \right]^{-1} \quad (7)$$

where $\mathbf{T}_{i,l}$, $\mathbf{T}_{i,d}$, $\mathbf{T}_{i,q}$, and $\mathbf{T}_{i,p}$ are the corresponding compliance transformation matrices (CTM), and detailed definitions can be found in [26] and [27].

1) *Output Compliance*: The global output system is identical to that of the double-parallelogram mechanism, namely, $O_d - x_d y_d z_d$. With respect to the output point O_d , the single-parallelogram (linkage-1) and PF mechanisms are serially connected, and then, they are further connected with the double-parallelogram mechanism (linkage-2) in parallel. Therefore, output compliance of the stage at point O_d can be obtained as

$$\mathbf{C}_o = \left[(\mathbf{T}_{p,d} \mathbf{C}_p \mathbf{T}_{p,d}^T + \mathbf{T}_{q,d} \mathbf{C}_q \mathbf{T}_{q,d}^T)^{-1} + \mathbf{C}_d^{-1} \right]^{-1} \quad (8)$$

where $\mathbf{T}_{p,d}$ and $\mathbf{T}_{q,d}$ are the CTMs from the local systems of the single-parallelogram and PF mechanisms to $O_d - x_d y_d z_d$, respectively.

2) *Input Compliance*: Although the actuation force is normally distributed on the armature polar surface, the input point can be assumed to be identical to O_p for calculation conducted in ideal conditions. Similar to the output compliance, the input compliance of the stage can be determined by

$$\mathbf{C}_{in} = \left[(\mathbf{T}_{d,p} \mathbf{C}_d \mathbf{T}_{d,p}^T + \mathbf{T}_{q,p} \mathbf{C}_q \mathbf{T}_{q,p}^T)^{-1} + \mathbf{C}_p^{-1} \right]^{-1} \quad (9)$$

where $\mathbf{T}_{d,p}$ and $\mathbf{T}_{q,p}$ are the CTMs from the local coordinate systems of the double-parallelogram and PF mechanisms to the system $O_p - x_p y_p z_p$, respectively.

3) *Motion Amplification Ratio*: Assume that the input and output motions are denoted by $\mathbf{x}_{in}$ and $\mathbf{x}_o$ in the global output coordinate system of the mechanism, i.e., $O_d - x_d y_d z_d$. Considering the inclined angle θ between the LSFHs in the PF mechanism and the driving direction, the force equilibrium relationship of the two LSFHs in the PF mechanism yields

$$\mathbf{C}_d^{-1} \mathbf{x}_o + (\mathbf{T}_{p,d} \mathbf{C}_p \mathbf{T}_{p,d}^T)^{-1} \mathbf{x}_{in} = \mathbf{C}_q^{-1} (\mathbf{x}_o - \mathbf{x}_{in}). \quad (10)$$

Accordingly, the matrix $\mathbf{J}$ relating the two motions $\mathbf{x}_o$ and $\mathbf{x}_{in}$ can be defined as

$$\mathbf{x}_o = \mathbf{J} \mathbf{x}_{in} \Rightarrow \mathbf{J} = (\mathbf{C}_q^{-1} - \mathbf{C}_d^{-1})^{-1} \left[(\mathbf{T}_{p,d} \mathbf{C}_p \mathbf{T}_{p,d}^T)^{-1} + \mathbf{C}_q^{-1} \right]. \quad (11)$$

By applying $\mathbf{x}_{in} = [1, 0, 0, 0, 0, 0]^T$, the motion amplification ratio can be determined by $A_r = \mathbf{x}_o(1)$ with $\mathbf{x}_o = \mathbf{J} \mathbf{x}_{in}$.

B. Dynamic Modeling of the Mechanism

Assume that u_1 is the generalized input coordinate along the actuation direction. The kinetic energy of the two parallelogram mechanisms subjecting to u_1 can be determined by

$$U_p = \sum_{i=1}^2 \left[\frac{1}{2} M_i \dot{u}_i^2 + \frac{1}{2} n_i I_i \left(\frac{\dot{u}_i}{l_i + 2R_i} \right)^2 \right] \quad (12)$$

where $u_2 = A_r u_1$, and M_1 and M_2 represent the moving mass of the input and output platforms, respectively. n_i is the number of linkages connected to the two platforms, and I_i , l_i , and R_i are the corresponding inertia moments, length of connecting beams, and notch radius for linkages, respectively.

With respect to the PF mechanism, its kinetic energy can be roughly estimated as

$$U_q = \frac{1}{8} m \dot{u}_1^2 (1 + A_r)^2 + \frac{1}{2} I \left(\frac{\dot{u}_1 \sin \theta}{l_3} \right)^2 \quad (13)$$

where l_3 , m , and I are the length, mass, and inertia moment of the two LSFHs in the PF mechanism, respectively.

Therefore, following the Lagrangian equation, the first natural frequency can be expressed by [28]

$$f_o = \frac{1}{2\pi} \sqrt{k_{in} \dot{u}_1 \left(\frac{\partial (U_p + U_q)}{\partial \dot{u}_1} \right)^{-1}} \quad (14)$$

where $k_{in} = \mathbf{K}_{in}(2, 2)$ with $\mathbf{K}_{in} = \mathbf{C}_{in}^{-1}$ being the input stiffness along the actuation direction.

IV. STRUCTURAL PARAMETER DETERMINATION

For nanopositioning stages, a good balance is essentially required between the two conflict performances, namely, motion stroke and natural frequency. With respect to the stage herein, the design criteria are set as follows: 1) a sufficient stroke with $s_o \geq \pm 75 \mu\text{m}$ and 2) a sufficient natural frequency around $f_o = 1 \text{ kHz}$ for the high bandwidth applications.

A. Structural Parameter Determination

To construct the magnetic circuit, the material perm-alloy 1J50 is selected for the armature and the stator, and a commercially available Nd-Fe-B PM sizing $5 \times 12.9 \times 18 \text{ mm}^3$ is employed to provide the dc bias flux, which has a remanence of $B_r = 1.18 \text{ T}$. With the coil winding, copper wire with a diameter of $d_w = 0.51 \text{ mm}$ is adopted with the capability of carrying a maximum current of $I_{\max} = 3.5 \text{ A}$. During design, a current of 3 A is adopted as the maximum to account for the modeling and manufacturing error. Although a smaller air gap d_0 will lead to a larger driving force, the working air gap herein is chosen as $d_0 = 0.2 \text{ mm}$ to account for manufacturing tolerances. The polar surface area of the armature is designed as the same as that of the PM, namely, $S_a = 5 \times 18 \text{ mm}^2$, so as to make the actuation force density as high as possible [22].

Considering saturation flux density $B_s(1.5 \text{ T})$ of perm-alloy 1J50, the number of required coil turns needs to satisfy

$$N \leq \frac{2d_0}{\mu_0 I} \left(B_s - \frac{\alpha B_r}{2} \cdot \frac{d_0 + x}{d_0} \right) \quad (15)$$

where the current and armature position are set as $I = 3 \text{ A}$ and $x = -80 \mu\text{m}$ to give a relatively large tolerance for design.

With further consideration of overall size of the magnetic circuit, the number of the coil turns is selected as $N = 70$, leading to an actuation force of $F_{am} = 35.68 \text{ N}$ at $x = 0$. Constrained by the magnetic circuit size, lengths of connecting beams in linkage-1, linkage-2, and LSFHs are, respectively, set as $l_1 = 10 \text{ mm}$, $l_2 = 15 \text{ mm}$, and $l_3 = 15.6 \text{ mm}$. The thicknesses for linkage-1 and linkage-2 are set as $b_1 = 2 \text{ mm}$ and $h = 28 \text{ mm}$, respectively. Recall the actuating force derived in (3); the constraint for input stiffness and amplification ratio A_r related to the desired stroke d_m can be expressed by

$$\frac{k_{in}}{A_r} \leq \frac{2B_a^2 S_a}{\mu_0 d_m} + \frac{B_a S_a}{\mu_0 d_m}. \quad (16)$$

TABLE I
PERFORMANCE OF THE DESIGNED STRUCTURE

Parameters	Values
Air gap d_0	0.2 mm
Polar surface, S_0	$5 \times 18 \text{ mm}^2$
Size of PM	$5 \times 12.9 \times 18 \text{ mm}^3$
Remanence of PM B_r	1.18 T
Leakage factor α	0.64
Coil turns N	70
Wire diameter d_w	0.51 mm
Maximum current $I_{\max}$	3.5 A
R_s/t_s for linkage-1	0.6 mm/0.3 mm
R_s/t_s for linkage-2	0.5 mm/0.2 mm
l_1, l_2, l_3	10 mm, 15 mm, 15.6 mm
Thickness b_1/h	2 mm, 28 mm
Angle θ	32.5°

TABLE II
PERFORMANCE OF THE DESIGNED STRUCTURE

	k_{in}	k_{out}	f_o	A_r	B_1	B_2	s_o
	N/ μm	N/ μm	Hz	—	T	T	μm
Anal	0.686	0.282	1017.2	1.286	1.02	0.300	85.3
FEA	0.621	0.266	996.2	1.42	1.018	0.302	104
Error	10.47%	6.01%	2.1%	9.44%	0.21%	0.59%	17.98%

Since k_{in} and A_r are determined by the flexure mechanism, the structural parameters are carefully adjusted through the trial-and-error method with the assistance of the analytical model, as established in Section III. The finally selected parameters for the stage are summarized in Table I.

B. Finite Element Verification

To demonstrate the designed stage, FEA-based numerical simulation is conducted through commercial software (ANSYS/Workbench). With the simulation, the electromagnetic property is investigated through the magnetostatic analysis modulus. Meanwhile, both static and modal analyses for the mechanical structure are conducted, and the obtained performances are then comparatively summarized in Table II, together with analytically calculated ones.

1) *Electromagnetic Simulation*: To verify the electromagnetic modeling, the distribution of the magnetic flux density is simulated and illustrated in Fig. 4 with the coil carrying a current of 3 A. As shown in Fig. 4, the obtained average flux density inside the two air gaps is estimated to be about 1.018 and 0.3 T, the density difference is caused by the flux bias provided by the PM. Referring to analytical flux densities obtained from (2), and the maximum deviation is checked to be less than 0.6%, as summarized in Table II, by taking FEA result as the benchmark. Combining the analytical force defined in (3), the maximum actuation force estimated by FEA is about $F_{am} = 34.03 \text{ N}$, showing a good agreement with the theoretical one (35.68N) as aforementioned. In addition, the maximum magnetic density is estimated to be 1.41 T ($< B_s = 1.5 \text{ T}$) under the maximum excitation condition, suggesting that the stage will work properly within the designed stroke.

2) *Mechanical Simulation*: By adopting an adaptive meshing strategy provided by the software, the mechanical model

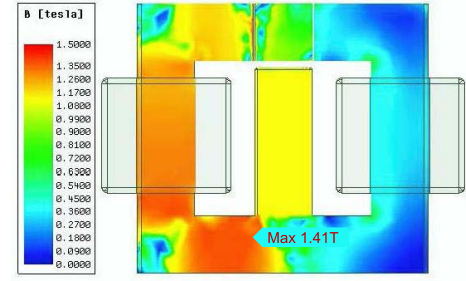


Fig. 4. Simulated distribution of the total magnetic flux density.

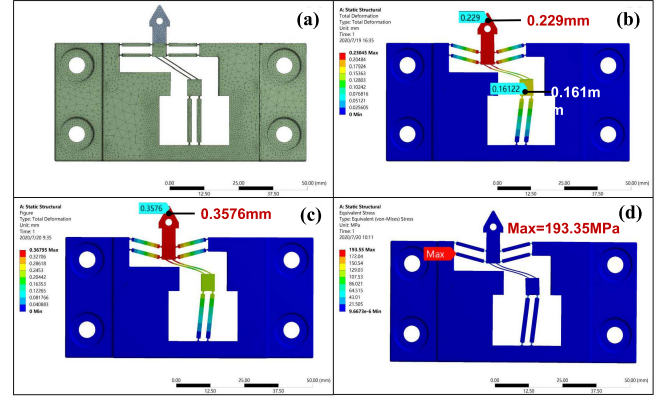


Fig. 5. Static analysis result of the mechanical structure: (a) meshed model, (b) deformation subjected to an input force of 100 N, (c) deformation subjected a force at the output end, and (d) equivalent Von Mises stress subjected to a maximum input force.

with finer meshing at a thinner place is obtained, as illustrated in Fig. 5(a). Through applying a constant force of 100 N on the polar surface of the armature, the resulting input motion and the output motion are presented in Fig. 5(b), leading to the assimilated amplification ratio of $A_r = 1.42$. In addition, by dividing the input motion by the input force, the input stiffness can be derived as $k_{in} = 0.621 \text{ N}/\mu\text{m}$. Similarly, by applying a force of 100 N on the output end, the output stiffness can be obtained from the deformation shown in Fig. 5(c) as $k_{out} = 0.266 \text{ N}/\mu\text{m}$. With respect to the static mechanical result, as summarized in Table II, the maximum deviation between the analytical and FEA result is less than 10.5%.

By applying the maximum actuation force on the input end of the mechanism, the distribution of the equivalent von-Mises stress is illustrated in Fig. 5(d), and the maximum one is identified to be around 194 MPa, which is well below the yield strength of the selected high-strength aluminum alloy ($> 300 \text{ MPa}$). Therefore, the repeatable elastic deformation can be guaranteed during the normal working process.

The FEA-based modal analysis is conducted for the mechanism with the assembled armature, and the first two resonant modes are shown in Fig. 6. The first-order mode in Fig. 6(a) is observed to agree well with the desired working motion, and a satisfied resonant frequency of about 996.2 Hz is achieved with a relative modeling error of 2.1%. Meanwhile, a twist of the armature and the end-effector is observed for the second-order mode in Fig. 6(b), which has a resonant frequency of

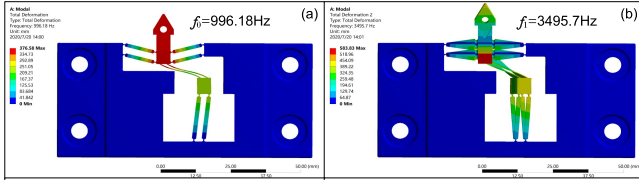


Fig. 6. Modal analysis result of the mechanism: (a) first mode and (b) second mode.

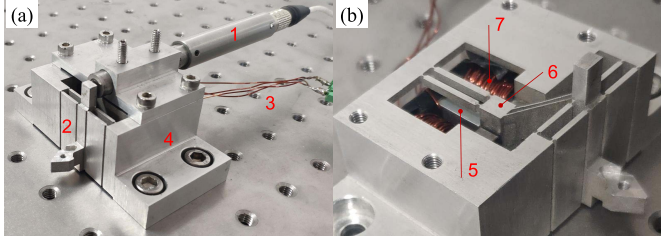


Fig. 7. Photography of the designed prototype: (a) overall structure and (b) inner structure, where: 1) the capacitive sensor probe; 2) flexure hinge; 3) coil wire; 4) base; 5) permanent magnetic; 6) armature; and 7) coil winding.

about 3.5 kHz. Since it is much larger than the first natural frequency, a stable motion along the desired direction can be guaranteed.

V. EXPERIMENTAL TESTING AND DISCUSSION

The prototype of the designed stage is photographically illustrated in Fig. 7(a), and the inner structure is further shown in Fig. 7(b) by removing the cover. A linear servo amplifier (SMA5005-1, Glentek Corporation, El Segundo, CA, USA) with a fixed gain of 0.8 A/V was employed to amplify the command signal for driving the coil winding, and a high bandwidth (20 kHz) capacitive displacement sensor with a measurement range of $\pm 250 \mu\text{m}$ (gauge 5810 with probe 5503, Microsense Corporation, Lowell, MA, USA) was employed to measure the practical displacement. A data acquisition board from National Instruments (PCI-6259) was used to send out the commands and collect the measured displacement, and both open- and closed-loop testings were conducted through the Desktop real-time modulus in MATLAB/Simulink with a sampling frequency of 20 kHz.

A. Basic Performance Testing

To investigate the motion range, a sinusoidal command with a maximum allowable amplitude of 4.375 V (3.5 A) and a low frequency of 1 Hz was adopted, and the resulting displacement related to the command is illustrated in Fig. 8. Overall, the displacement of the stage ranges from -88.5 to $101.5 \mu\text{m}$, and a clear hysteresis loop is observed with a maximum value of about $21 \mu\text{m}$, which is about 11% of the full motion span.

A sweep excitation using a harmonic command with a constant amplitude of 0.1875 V and a linearly varying frequency ranging from 1 Hz to 1 kHz was conducted to obtain the frequency response feature. By employing Fourier transformation of both input command and output response, the resulting frequency response function is illustrated in Fig. 9. From the

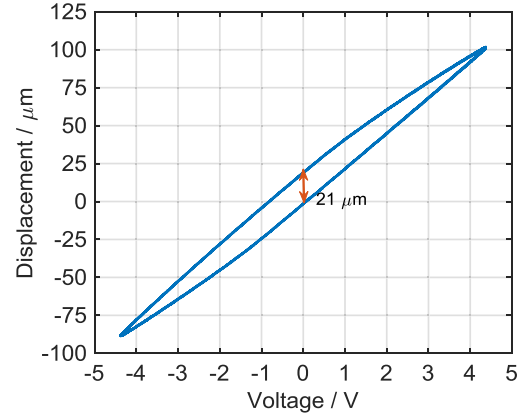


Fig. 8. Relationship between the command voltage and output motion.

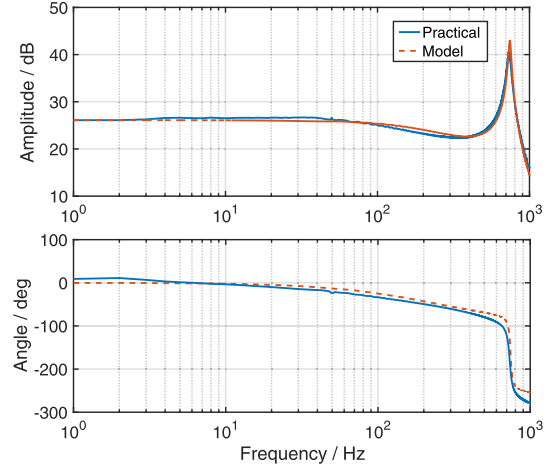


Fig. 9. Bode diagram of the open-loop system.

resonant peak in Fig. 9, the first natural frequency is identified to be about 743 Hz. Comparing with the estimated result (996.2 Hz), this practical resonant frequency is about 25% smaller, which might be jointly caused by manufacturing error-induced stiffness decrease and an extra mass of the block for displacement sensing, as shown in Fig. 7.

Considering the second-order nature of the mechanical structure and the first-order nature of the driving circuit, the whole system can be described as a typical third-order system. Accordingly, by identifying the static amplitude, resonant frequency, and frequency-dependent phase-lag from the tested frequency response function, the nominal transfer function of the stage can be obtained as

$$P_m(s) = \frac{7.5315 \times 10^{11}}{(s + 1382)(s^2 + 186.7s + 2.179 \times 10^7)} \quad (17)$$

where s is the Laplace operator.

As comparatively demonstrated in Fig. 9, the estimated frequency response function from $P_m(s)$ agrees well with the practically measured one, suggesting that the identified system model $P_m(s)$ is reliable for tuning controller parameters.

B. Positioning Performance Testing

The schematic of the constructed control system is illustrated in Fig. 10. A typical proportional—integral—differential

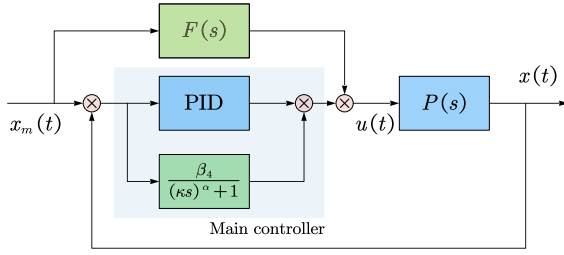


Fig. 10. Schematic of the constructed control system, where $P(s)$ denotes the stage to be controlled with a nominal transfer function of $P_m(s)$.

(PID) controller with a parallel fractional order filter is employed to serve as the main controller for the stage, which can be expressed by

$$C(s) = \beta_1 + \frac{\beta_2}{s} + \frac{\beta_3 N_s s}{s + N_s} + \frac{\beta_4}{(\kappa s)^\alpha + 1} \quad (18)$$

where β_i , $i = 1, 2, 3, 4$ are control parameters, N_s and κ are, respectively, the sampling frequency and fractional order gain, and $\alpha \in (0, 1)$ is a differential order.

To improve dynamic response behavior, a system dynamics inversion-based feedforward compensator $F(s)$ is further adopted, which is

$$F(s) = P_m(s)^{-1} Q(s) \quad (19)$$

where $Q(s)$ is a low-pass filter. Considering the third-order nature of the identified system model $P_m(s)$, a third-order Butterworth low-pass filter is adopted herein.

1) *Parameter Tuning for the Main Controller*: From the loop-shaping perspective, controller parameters can be tuned by minimizing the difference between the frequency response functions of the open-loop control system and the desired system transfer function $L_d(s)$. Considering the ideal shape for the frequency response, Bode's ideal transfer function using a fractional order integrator is employed, which can be expressed by [29]

$$L_d(s) = \left(\frac{\omega_c}{s}\right)^\gamma, \quad \gamma \in \mathbb{R}^+ \quad (20)$$

where ω_c is the gain crossover frequency and $\gamma \in (1, 2)$ is the integrator order.

To guarantee system robustness, a sufficient modulus margin M_m is required, which can be satisfied by setting points of $L_s(s)$ in the Nyquist curve to be below a specified line d_k , which is tangential to a circle at the critical point with a radius of M_m and orthogonal to the line connecting $L_d(j\omega_k)$ and the critical point [30].

By defining the point on the line d_k as $\mathcal{F}(x + jy, d_k) = 0$, controller parameters can be optimized by solving the following minimization problem [30]:

$$\min \sum_{i=1}^m \sum_{k=1}^K |C(j\omega_{i,k}) P_m(j\omega_{i,k}) - L_d(j\omega_{i,k})|^2 \quad (21)$$

$$\text{s.t. } \mathcal{F}(L_s(j\omega_{i,k}), d_{i,k}) < 0 \quad \forall i = 1, \dots, m \quad \forall k = 1, \dots, K. \quad (22)$$

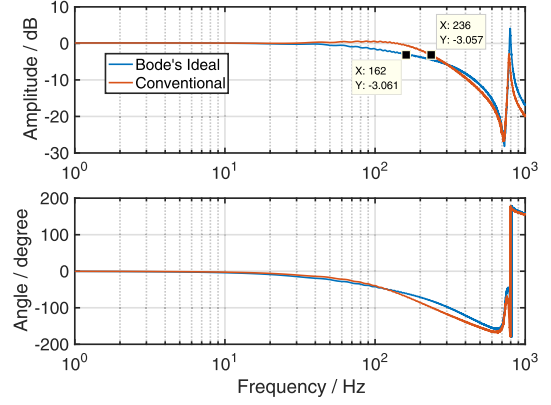


Fig. 11. Frequency response function of the closed-loop system.

Following the relationship shown in Section B, the fractional order γ for Bode's ideal transfer function is directly set as $\gamma = 1.3$ constrained by $PM > (\pi/3)$ and $\zeta > 0.7$. Meanwhile, the order for the fractional order filter is set as $\alpha = 0.1$, and frequency ω_c is derived as $\omega_c = 1144$ rad/s by setting rise time $T_r = 1.2$ ms. Accordingly, the optimized main controller using Commande Robuste d'Ordre Non Entier (CRONE) approximation is

$$C_1(s) = \frac{0.0967(s + 1416)(s^2 + 288.5s + 2.081 \times 10^7)}{s(s + 2807)(s + 2 \times 10^4)}. \quad (23)$$

Similarly, by setting $\gamma = 1$, the tuning process changes to be a normal loop-shaping method, leading to an optimized controller as

$$C_2(s) = \frac{0.1284(s + 734.3)(s^2 + 256.5s + 2.078 \times 10^7)}{s(s + 2807)(s + 2 \times 10^4)}. \quad (24)$$

2) *Practical Performance of the Designed Control System*: Through sweep testing of the practically formed closed-loop system, the frequency response functions for the two control systems are illustrated in Fig. 11. As shown in Fig. 11, a bandwidth of about 162 Hz (-3 dB) and a resonant amplitude near 5 dB are observed for the main control system tuned without using Bode's ideal transfer function. By contrast, taking advantage of Bode's ideal transfer function, the tuned controller $C_1(s)$ may lead to a higher bandwidth of 236 Hz (-3 dB) and a smaller resonant vibration less than 0 dB, demonstrating the feasibility and outperforming property of the designed main controller.

To investigate trajectory tracking performance of the developed nanopositioning stage, practical tracking of a harmonic trajectory with an amplitude of $20 \mu\text{m}$ and a frequency of 20 Hz is conducted using $C_1(s)$ as the main controller. The tracking performance and following error obtained with and without feedforward compensator $F(s)$ are illustrated in Fig. 12. As shown in Fig. 12(a), a relatively large phase-lag is observed for the system without adding a feedforward compensator, which may lead to a relatively large following

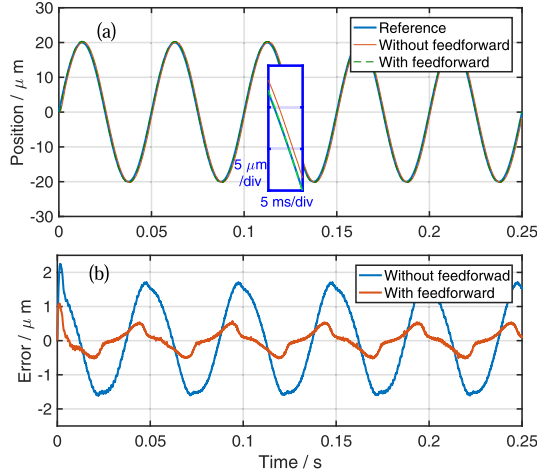


Fig. 12. Harmonic trajectory tracking: (a) tracking result and (b) tracking error.

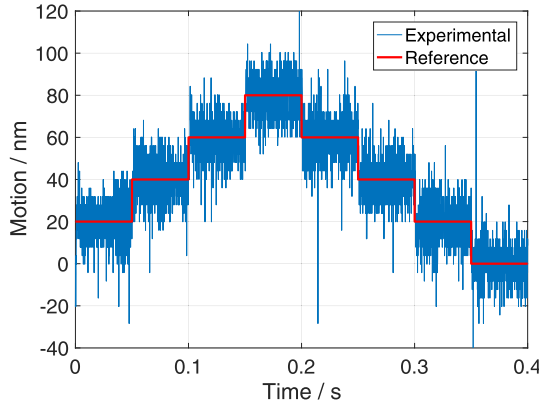


Fig. 13. Stair tracking for motion resolution identification.

error around $\pm 1.8 \mu\text{m}$. Taking advantage of the compensator $F(s)$, the following error is reduced to be about $\pm 0.4 \mu\text{m}$, which is about $\pm 1\%$ of the full motion span ($40 \mu\text{m}$).

Furthermore, to demonstrate the nanopositioning capability of the stage, a stair tracking testing with each step of 20 nm and time interval of 50 ms is conducted, and the result is shown in Fig. 13. A clear stair response dominated by electromagnetic noises is observed, suggesting that a motion resolution of at least 20 nm can be achieved for the stage.

C. Performance Comparison

Taking the stroke s_o and natural frequency f_o as the basic performance measurement, the performances of the designed stage are comparatively presented in Table III with the reported state-of-the-art NSEA-based stages. As shown in Table III, the reported NSEA-based stages are mainly designed to achieve short-stroke motions in tens of micrometers or long-stroke motions in the millimeter scale. Comparing with the motion in hundreds of micrometers, the designed one presented in this work shows an outperforming resonant frequency compared with one presented in [5]. In addition, the designed stage is also compared with certain typical piezo-actuated stages in

TABLE III
PERFORMANCE COMPARISON

	Normal-stressed electromagnetic					Piezo-actuated			
	This work	[18]	[31]	[5]	[32]	[19]	[33]	P-621.1	P-750
$s_o / \mu\text{m}$	190	30	35.5	170	1500	2000	288.3	120	75
f_o / Hz	743	3000	4780	210	122	178	155	800	600

Table III, including the commercial ones from the PI Corporation, Karlsruhe, Germany (P-621.1 & P-750). It suggests that the designed stage can have a much higher resonant frequency when the stroke is similar to that of the piezo-actuated stages.

VI. CONCLUSION

In this article, a novel compliant nanopositioning stage is developed by employing a self-designed NSEA, featuring the high force density, large motion range, and linear actuation behavior. Through analytical modeling and finite element analysis of both the electromagnetic and mechanical components, the structural parameters for the stage are deliberately selected and validated to satisfy the design requirements. The stroke of the prototype is examined to be around $\pm 95 \mu\text{m}$ with a maximum hysteresis around 11% of the motion span, and the first natural frequency is identified to be 743 Hz .

A control system for the stage is constructed by combining a PID-based main controller and a system dynamics inversion-based feedforward compensator. Taking advantage of the identified system model, Bode's ideal transfer function-based loop-shaping method is employed to tune the parameters for the main controller. Finally, the following error is tested to be less than $\pm 1\%$ for tracking a harmonic with a frequency of 20 Hz , and a fast and accurate stair tracking with each step of 20 nm is well demonstrated.

APPENDIX

A. Approximation of Fractional Order Calculus

The CRONE approximation is adopted to calculate the fractional calculus. Taking s^α for example, it can be approximated within a frequency range $[\omega_l, \omega_h]$ by [34], [35]

$$s^\alpha = \varepsilon \prod_{n=1}^N \frac{1 + \frac{s}{\omega_{z,n}}}{1 + \frac{s}{\omega_{p,n}}}, \quad \alpha > 0 \quad (\text{A1})$$

where ε is used to adjust a zero gain for the frequency of 1 rad/s , and variables $\omega_{p,n}$ and $\omega_{z,n}$ are defined as [34], [35]

$$\omega_{p,n} = \omega_{z,n-1} \left(\frac{\omega_h}{\omega_l} \right)^{\frac{\alpha}{N}} \quad (\text{A2})$$

$$\omega_{z,n} = \omega_{p,n-1} \left(\frac{\omega_h}{\omega_l} \right)^{\frac{1-\alpha}{N}} \quad (\text{A3})$$

$$\omega_{z,1} = \omega_l \left(\frac{\omega_h}{\omega_l} \right)^{\frac{1-\alpha}{2N}}. \quad (\text{A4})$$

Practically, to avoid high order differential calculation, the CRONE approximation order is set as $N = 1$ for this study.

B. Property of Bode's Ideal Transfer Function

Bode's ideal transfer function has an isodamping property around the frequency ω_c , and the output of the corresponding closed-loop system to a unit step input $R(s) = \frac{1}{s}$ can be obtained in the time domain as [29]

$$y_{sp}(t) = \mathcal{L}^{-1} \left\{ \frac{\omega_c^\gamma}{s(s^\gamma + \omega_c^\gamma)} \right\} = 1 - \sum_{n=0}^{\infty} \frac{(-\omega_c^\gamma t^\gamma)^n}{\Gamma(1 + \gamma n)} \quad (B1)$$

where $\Gamma(\cdot)$ is the Gamma function and the equivalent damping ratio ξ is [29]

$$\xi = \cos \left(\pi - \frac{\pi}{\gamma} \right). \quad (B2)$$

With respect to a given order γ , the relationship between the cutoff frequency ω_c and the rise time T_r , which is defined as the time for the response to evolve from 10% to 90% of the final step value, is [29]

$$\omega_c = \frac{0.131(\gamma + 1.157)^2}{(\gamma - 0.724)T_r}. \quad (B3)$$

The gain margin for this ideal system $L_d(s)$ is theoretically infinite, and a fruitful phase margin larger than $(\pi/3)$ is desired [29]

$$PM = \pi \left(1 - \frac{\gamma}{2} \right) > \frac{\pi}{3}. \quad (B4)$$

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